

CLUSTERING

CLUSTERING ALGORITHMS – PART V

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CLUSTERING ALGORITHMS

The following concepts will be introduced:

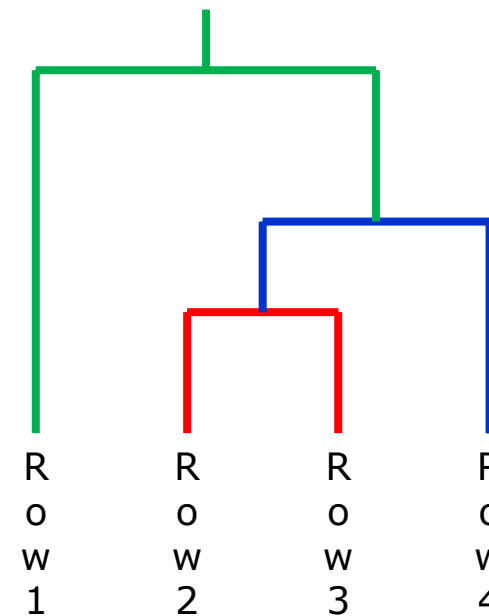
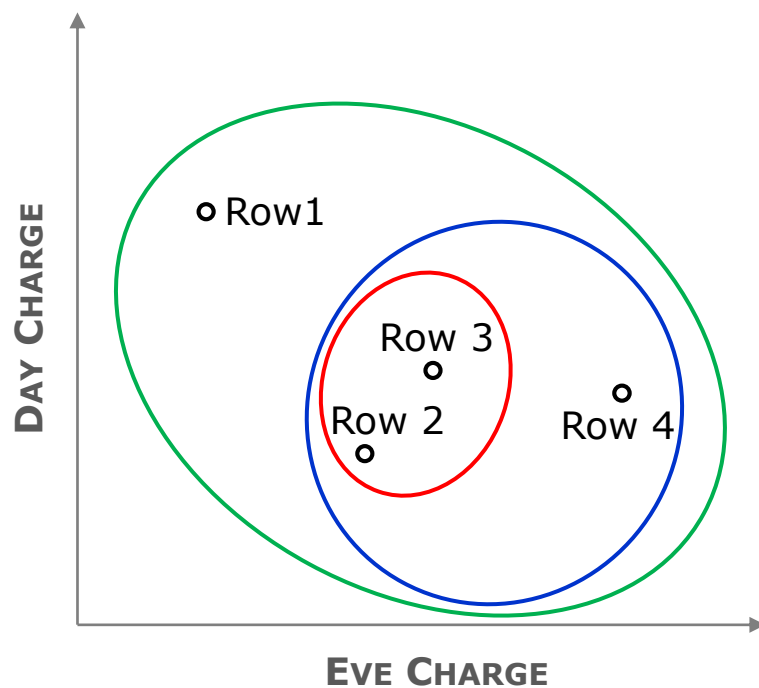
- ✓ **HIERARCHICAL CLUSTERING**
- ✓ **AGGLOMERATIVE/DIVISIVE HIERARCHICAL CLUSTERING**
- ✓ **DENDROGRAM**
- ✓ **SINGLE/COMPLETE/AVERAGE LINKAGE**
- ✓ **WARD'S METHOD**

HIERARCHICAL CLUSTERING can be

- ✓ **AGGLOMERATIVE;** starts with **all objects** (records) **as individual clusters** and, at each step, **merges the closest pair of clusters**. This requires defining a **notion of cluster proximity**
- ✓ **DIVISIVE;** starts with one, **all-inclusive cluster** and, at each step, **splits a cluster until only singleton clusters** of individual objects remain. In this case, we need to decide **which cluster to split** at each step and **how to do the splitting**

AGGLOMERATIVE HIERARCHICAL CLUSTERING TECHNIQUES are by far the most common, and we focus in this lecture exclusively on these methods.

HIERARCHICAL CLUSTERING is often displayed graphically using a tree-like diagram called **DENDROGRAM**, which displays both the cluster-subcluster relationships and the order in which the clusters were merged (agglomerative) or split (divisive)



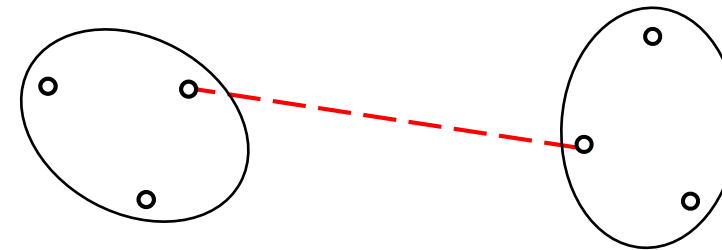
AGGLOMERATIVE HIERARCHICAL CLUSTERING ALGORITHM

1. Compute the proximity matrix, if necessary
2. **REPEAT**
3. Merge the closest two clusters
4. Update the proximity matrix to reflect the proximity between the new cluster and the original clusters
5. **UNTIL** only one cluster remains



HOW TO COMPUTE THE PROXIMITY BETWEEN TWO CLUSTERS?

HOW TO DEFINE CLUSTER PROXIMITY?



MIN OR SINGLE LINKAGE

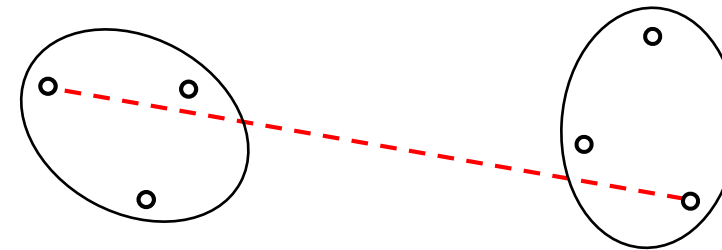
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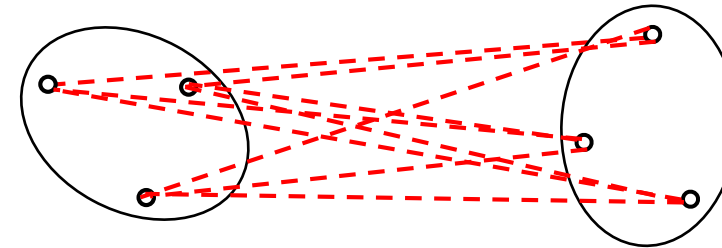
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GROUP AVERAGE OR AVERAGE LINKAGE

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HOW TO DEFINE CLUSTER PROXIMITY?

An alternative technique, **WARD'S METHOD**, assumes that a cluster is represented by its centroid, but it measures the proximity between two clusters in terms of the increase in the Sum of Squared Errors that results from merging two clusters.

THE KEY ISSUES OF AGGLOMERATIVE HIERARCHICAL CLUSTERING are the following:

- ✓ **LACK OF A GLOBAL OBJECTIVE FUNCTION;** agglomerative hierarchical clustering techniques use various criteria to decide locally, at each step which clusters should be merged. This avoids the difficulty to solve a hard combinatorial optimization problem
- ✓ **ABILITY TO HANDLE DIFFERENT CLUSTER SIZES;** allows to treat the relative sizes of the pairs of clusters that are merged (this applies to Ward and group average). The possible approaches are weighted (clusters are treated equally) and unweighted (takes the number of objects in each cluster into account)
- ✓ **MERGING DECISIONS ARE FINAL;** once a decision is made to merge two clusters, it can not be undone at a later time. This prevents a local optimization criterion to become a global optimization criterion

WEAKNESSES OF AGGLOMERATIVE HIERARCHICAL CLUSTERING are the following:

- ✓ **COMPUTATIONALLY AND SPACE INTENSIVE;** agglomerative hierarchical clustering algorithms are expensive in terms of their computational and storage requirements
- ✓ **MERGING DECISIONS ARE FINAL;** can cause trouble for noisy, high-dimensional data such as document data